LEAST SQUARES IDENTIFICATION OF AN LED AND LDR

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1.0 Learning Outcomes

Upon completing this practical exercise, students should be able to demonstrate the Learning Outcomes presented in Table 1. At the end of the exercise, revisit each Learning Outcome and confirm that it has been achieved.

Table 1: *Learning Outcomes for the Practical Exercise*

ID	Learning Outcome	Achieved
1	Log input and output data from the LED and light-dependent resistor (LDR), and arrange it as a linear least squares problem.	☐
2	Identify a static polynomial curve and a first-order dynamic model of the LED-to-LDR system with linear least squares.	☐
3	Validate each identified model on a separate data set and judge how well it represents the system.	☐

2.0 Hardware Set-Up

This practical exercise builds on the hardware set-up described in the practical exercises, LED (PWM) and LDR (Light Dependent Resistor), and uses the LED and LDR arrangement inside a light shield described in the practical exercise, Light-Level Control with an LED and LDR. The set-up is not repeated here. Only the Base + Sense kit is needed.

3.0 Practical Investigation via Simulink

MATLAB and Simulink should be used to complete this practical exercise. Linear least squares is used to identify two models of the LED-to-LDR system: a static curve relating the PWM value to the steady LDR reading, and a first-order dynamic model describing how quickly the reading responds. The exercise applies the method of the practical exercise, Parameter Estimation of a DC Motor using Linear Least Squares, to a second system, and supports the ACE-CORE learning guide, Linear Least Squares for Parameter Estimation.

Background: Linear in the Parameters

Least squares can fit any model that is linear in its unknown parameters θ, even when it is nonlinear in the input. Stacking one equation for each data point gives $= \Phi\theta$, which is solved by:

$$\hat{\theta} = (\Phi^T \Phi)^{-1} \Phi^T y$$

Two models are identified in this exercise. The static model is a quadratic in the PWM value u, and the dynamic model is a first-order difference equation:

$$\text{Static}: \quad y = \theta_0 + \theta_1 u + \theta_2 u^2 \qquad \text{Dynamic}: \quad y[k] = -a_1\, y[k-1] + b_0\, u[k-1]$$

Identify the Static Curve

[Screenshot to insert] *Signal source (staircase or random steps, PWM 0–255) → PWM output (LED); Analog Input (LDR) → Scope; u, y and t logged with Save format Structure With Time.*

Figure 1: *Simulink Model for Logging the LED Input and LDR Output*

1. Build the model in Figure 1 with a sample time of 0.02 s, and log the PWM value, the LDR reading and the time as *u*, *y* and *t*.
2. Apply a staircase from PWM 0 to 255 in 15 levels, holding each for 2 s, using the staircase function from the practical exercise, PWM to Speed Characteristic of a DC Motor.
3. Average the LDR reading over the last second of each level, and fit both a straight line and a quadratic with the script below.

MATLAB Script: identifying the static curve with least squares

```
%% u_lev and y_lev: PWM levels and averaged LDR readings (column vectors)
Phi1 = [ones(size(u_lev)) u_lev];                 % straight line
Phi2 = [ones(size(u_lev)) u_lev u_lev.^2];        % quadratic
th1 = (Phi1'*Phi1) \ (Phi1'*y_lev);
th2 = (Phi2'*Phi2) \ (Phi2'*y_lev);
e1 = y_lev - Phi1*th1;   e2 = y_lev - Phi2*th2;
fprintf('RMS error: line %.1f, quadratic %.1f counts\n', rms(e1), rms(e2));
uu = linspace(0,255,200)';
plot(u_lev, y_lev, 'ko', uu, [ones(200,1) uu]*th1, 'r--', ...
     uu, [ones(200,1) uu uu.^2]*th2, 'b'); grid on;
xlabel('PWM value, u'); ylabel('LDR reading, y [counts]');
legend('Measured', 'Straight line', 'Quadratic', 'Location', 'best');
```

Identify the Dynamic Model

1. Apply a sequence of random PWM steps between 60 and 200, each lasting between 0.1 s and 1 s, for 30 s. The MATLAB function `randi` can generate the levels and durations in advance.

2. Use the first 20 s of data to estimate a_1 and b_0 with the script below, and the last 10 s to validate the model.
3. Convert a_1 into a time constant, $\tau = -T_s/\ln(-a_1)$, and record the results in Table 2.
4. Repeat the test around a low PWM level (40 to 100) and a high level (160 to 240), and compare the time constants.

MATLAB Script: identifying and validating a first-order dynamic model

```
u_k = u.signals.values(:);              % PWM value
y_k = y.signals.values(:);              % LDR reading [counts]
t_k = t.signals.values(:);              % time [s]
Ts = 0.02;
est = t_k < 20;  val = ~est;            % estimation / validation data

ue = u_k(est) - mean(u_k(est));
ye = y_k(est) - mean(y_k(est));
Phi = [-ye(1:end-1), ue(1:end-1)];
theta = (Phi'*Phi) \ (Phi'*ye(2:end));          % theta = [a1; b0]
a1 = theta(1);  b0 = theta(2);
tau = -Ts/log(-a1)                              % time constant [s]

%% Validate: simulate the model from the input alone on the unused data
uv = u_k(val) - mean(u_k(est));  yv = y_k(val) - mean(y_k(est));
ys = zeros(size(yv));
for k = 2:length(yv), ys(k) = -a1*ys(k-1) + b0*uv(k-1); end
fit = 100*(1 - norm(yv - ys)/norm(yv - mean(yv)))   % fit [%]
```

The means are removed before fitting because the dynamic model describes changes about an operating point, not the absolute LDR reading. A fit above about 80% shows a model that captures the main behaviour. LDRs are typically slower to respond when the light decreases than when it increases, and slower at low light levels, so a single first-order model may fit some operating ranges better than others.

Table 2: *Identified Models for Your LED and LDR*

Model	Parameters	RMS error or fit	Time constant, τ (s)
Static, straight line			–
Static, quadratic			–
Dynamic, PWM 60–200			
Dynamic, PWM 40–100			
Dynamic, PWM 160–240			

1. Why must the regressors in Φ be linear in the parameters, but not necessarily in the input?

2. Why is the model validated on data that was not used to estimate it?
3. How would you change the method if the LDR responded at different speeds to rising and falling light?

4.0 Concluding Remarks

This practical exercise used linear least squares to identify static and dynamic models of the LED-to-LDR system. Using MATLAB, Simulink and ACE-Lab hardware, a quadratic curve was fitted to the steady readings and a first-order difference equation to the response to random steps, and each model was validated on separate data.

The quadratic described the nonlinear static curve better than a straight line, showing that least squares is not limited to linear relationships. The dynamic model captured the main behaviour around each operating point, while changes in the time constant with light level showed where a single linear model reaches its limits.

Recommended Citation

If you have used this practical exercise in your teaching, learning or research, please cite it using the reference below, adapting the format as required by your preferred referencing style.

Pickering, J. E. (2026). Least Squares Identification of an LED and LDR. ACE-Lab. Teaching Notes: ACE-Lab Practical Exercises, Version 1.0. Zenodo. https://zenodo.org/records/23725330